Apple iTunes Business Analysis

By

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For

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1. Executive Summary

This report presents the findings derived from the analysis of the Apple's iTunes store datasets. It presents insights about sales performance, customer behaviour, employee productivity, and revenue trends over time. The analysis is based on 11 datasets, that contained details of 59 customers, 3503 tracks, 275 artists, 347 albums, and 25 genres within the calendar years of 2017 and 2020.

Key findings from the analysis:

- Total revenue generated across all years was found to be $4,709.43.
- The best performing year was found to be 2019 with a total revenue of $1,221.66.
- Most popular genre was found to be Rock which was 55.39% of all sales.
- Most popular artist was found to be Queen with 192 tracks sold.
- Most popular track was found to be War Pigs by Cake with 31 purchases.
- Best performing employee was found to be Jane Peacock with $1,731.51 in revenue.
- Highest spending customer was found to be Frantisek Wichterlova from Czech Republic with a total purchase for the amount of $144.54.
- The top performing region was found to be Prague, Czech Republic with total revenue of $273.24.
- All customers were found to be repeat buyers with an average of approximately 10 purchases each.

2. Data Cleaning

Before drawing insights from the datasets, all 11 datasets were analysed for nulls, and the potential primary key columns were checked for duplicates.

2.1 Issues that were identified and resolved

- Checked for nulls and empty strings in all tables and found:

Table Name	Issue	Resolution
Customer	The column company had 49 nulls	These columns do not affect the ability to pull insights to answer the business questions. Therefore, the null rows were not deleted.
	The column state had 29 nulls	
	The column postal_code had 4 nulls	
	The column phone had 1 null	
	The column fax had 47 nulls	
Employee	The column reports_to had 1 null	Manually inserted the dataset again and fixed the error.
Invoice_line	The column billing_state had 304 nulls	These columns do not affect the ability to pull insights to answer the business questions. Therefore, the null rows were not deleted.
	The column billing_postal_code had 55 nulls	
Track	The column composer had 978 nulls	This column does not affect the ability to pull insights to answer the business questions and therefore were not deleted.

- The nulls that were found in critical columns were fixed and the rest were kept for analysis,
- No duplicate keys were found in the datasets.

3. Analysis

3.1 The Most and Least Popular Genre:

Most Popular Genres	Tracks Sold	Total Revenue
Rock	2635	$2,608.65
Metal	619	$612.81
Alternative & Punk	492	$487.08
Latin	167	$165.33
R&B/Soul	159	$157.41

Least Popular Genre	Tracks Sold	Total Revenue
Hip Hop/Rap	33	$32.67
Heavy Metal	8	$7.92
Soundtrack	5	$4.95
TV Shows	2	$1.98
Drama	1	$0.99

Most popular music genre was found to be Rock with a total of 2635 tracks sold making a revenue of $2608.65 while the least popular music genre was found to be drama with a total of 1 track sold and a revenue generated of $0.99.

3.2 Most and Least Popular Genre by Country:

Rock Is the most popular genre in 23 out of the 24 countries. Argentina was the only country which preferred Alternative & Punk with 42.50% of the sales being in that genre. Sweden was found to have the most percentage of Rock purchases with 78.95% of the sales being for this genre.

3.3 Tracks Per Genre vs Sales Correlation

Genre	Catalog %	Sales %	Sales/Track
Rock	26.12	55.39	2.88
Alternative	**0.97**	**2.46**	**3.44 (MOST EFFICIENT)**
Easy Listening	0.69	1.56	3.08
R&B/Soul	1.57	3.34	2.89
Latin	3.4	3.51	1.4
Heavy Metal	**0.2**	**0.17**	**1.14 (LEAST EFFCIENT)**

3.4 Average Price Per Track Per Genre

The genres Drama and TV Shows were priced higher at $1.99 per track while the other genres were uniformly priced at $0.99. This confirmed that volume rather than pricing drives the revenue across different genres.

4. Track and Artist Analysis

4.1 Most and Least Popular Tracks

Category	Track	Artist	Purchases	Revenue
Most Popular	**War Pigs**	**Cake**	**31**	**$30.69**
2nd Most Popular	**Highway Chile**	**Jimi Hendrix**	**14**	**$13.86**
3rd Most Popular	**Are You Experienced?**	**Jimi Hendrix**	**14**	**$13.86**
Least Popular (tied)	**705 tracks**	**Various**	**1 each**	**$0.99 each**

4.2 Tracks Never Purchased

There were a total of 1697 tracks that were never purchased which made them the least popular.

4.3 Top 10 Popular Artists

Artist	Tracks Sold	Total Revenue
Queen	**192**	**$190.08**
Jimi Hendrix	187	$185.13
Nirvana	130	$128.7
Red Hot Chili Peppers	130	$128.7
Pearl Jam	129	$127.71
Guns N' Roses	124	$122.76
AC/DC	124	$122.76
Foo Fighters	121	$119.79
The Rolling Stones	117	$115.83
Metallica	106	$104.94

- 14 artists sold 1 track each and the majority of them were classical and Brazilian music artists.

4.4 Top 10 Rock Bands by Tracks Produced

Artist Name	Total Tracks
Led Zeppelin	**114**
U2	112
Deep Purple	92
Iron Maiden	81
Pearl Jam	54
Van Halen	52
Queen	45
The Rolling Stones	41
Creedence Clearwater Revival	40
Kiss	35

4.5 Most Common Tracks Combinations

Track 1	Artist 1	Track 2	Artist 2	Times Purchased Together	Purchase %
Love Or Confusion	Jimi Hendrix	I Don't Live Today	Jimi Hendrix	9	1.47
May This Be Love	Jimi Hendrix	Highway Chile	Jimi Hendrix	9	1.47
Foxy Lady	Jimi Hendrix	Third Stone From The Sun	Jimi Hendrix	9	1.47
Third Stone From The Sun	Jimi Hendrix	Remember	Jimi Hendrix	8	1.3
Can You See Me	Jimi Hendrix	Remember	Jimi Hendrix	8	1.3

- Tracks by Jimi Hendrix dominate all purchase combinations with customers frequently buying multiple Hendrix tracks in the same transaction.

4.6 Songs Longer Than Average

- The average song length was found to be 3,93,599 milliseconds (6.54 minutes). A total of 494 tracks (14.1%) were found to be longer than average.
- The longest track was found to be Occupation/Precipice which was 88.12 minutes long.

4.7 Most Popular Albums and Playlists

Album	Artist	Times Purchased	Total Revenue
Are You Experienced?	Jimi Hendrix	187	$185.13
Faceless	Godsmack	96	$95.04
Mezmerize	System Of A Down	93	$92.07
Get Born	JET	90	$89.1
The Doors	The Doors	83	$82.17

5. Revenue & Sales Trends

5.1 Overall Revenue

Metric	Value
Total Revenue (2017-2020)	$4,709.43
Total Invoices	614
Average Invoice Value	$7.67
Minimum Invoice Value	$0.99
Maximum Invoice Value	$23.76

5.2 Yearly Revenue Trend

Year	Total Invoices	Total Revenue	Avg Invoice Value	Year on Year Change
2017	154	$1,201.86	7.8	Baseline
2018	151	$1,147.41	7.6	-$54.45 (-4.5%)
2019	159	$1,221.66	7.68	+$74.25 (+6.5%)
2020	150	$1,138.5	7.59	-$83.16 (-6.8%)

5.3 Quarterly Revenue Trend

Quarter	Avg Revenue	Performance
Q1 (Jan-Mar)	**$327.44**	**Best Quarter**
Q3 (Jul-Sep)	$301.95	2nd Best
Q2 (Apr-Jun)	$297.74	3rd
Q4 (Oct-Dec)	**$250.22**	**Weakest Quarter**

5.4 Monthly Revenue Trend

Month Name	Total Revenue	Performance
March	**$456.39**	**Peak Month**
April	$442.53	2nd
January	$438.57	3rd
August	$426.69	4th
November	**$291.06**	**Lowest Month**

5.5 Revenue by Media Type

Media Type	Total Purchases	Total Revenue
MPEG audio file	**4259**	**$4216.41**
Protected AAC audio file	439	$434.61
Purchased AAC audio file	35	$34.65
AAC audio file	21	$20.79
Protected MPEG-4 video file	3	$2.97

6. Customer Analysis

6.1 Top Spending Customers

Customer	Country	Invoices	Total Spent	Avg Invoice
Frantisek Wichterlova	Czech Republic	18	$144.54	$8.03
Helena Holy	Czech Republic	12	$128.7	$10.72
Hugh O'Reilly	Ireland	13	$114.84	$8.83
Manoj Pareek	India	13	$111.87	$8.61
Luis Goncalves	Brazil	13	$108.9	$8.38
Fernanda Ramos	Brazil	15	$106.92	$7.13
Joao Fernandes	Portugal	13	$102.96	$7.92
Wyatt Girard	France	11	$99.99	$9.09
Francois Tremblay	Canada	9	$99.99	$11.11
Jack Smith	USA	12	$98.01	$8.17

6.2 Customer Lifetime Value

Metric	Value
Average Customer Lifetime Value	$79.82
Highest Customer Value	$144.54
Lowest Customer Value	$29.7
Average Invoices per Customer	$10.41
Average Invoice Value	$7.76
Average Customer Lifespan	1,244 days (~3.4 years)

6.3 Repeat vs One Time Customers

No customer has made a single purchase and all 100% of the 59 customers were repeat buyers. The most common purchase frequency is 9-10 invoices per customer, accounting for 40% of the total customer base.

6.4 Revenue per Customer by Country

Rank	Country	Total Customers	Revenue/Customer
1	**Czech Republic**	**2**	**$136.62**
2	Ireland	1	$114.84
3	Spain	1	$98.01
4	Chile	1	$97.02
5	Portugal	2	$92.57
Last	Denmark	1	$37.62

6.5 Inactive Customers

Based on the dataset, it was found that there were 16 customers that were inactive with no purchases for more than 6 months. This shows that 27% of the customers have been inactive for more than 6 months. The most inactive customer is Daan Peeters from Belgium (466 days). High value inactive customers include Francois Tremblay, Luis Rojas, and Enrique Munoz high priority targets for reengagement.

6.6 Average Time Between Purchases

Metric	Value
Overall Average	140 days
Most Frequent Buyer	Frantisek Wichterlova (74 days)
Least Frequent Buyer	Robert Brown (392 days)

6.7 Genre Diversity

- All customer purchases are from more than 1 genre.
- High variety buyers were found to be 91.53% who bought from 7 or more different genre.
- High variety buyers spend more on average ($81.38) than medium diversity buyers ($62.96).

7. Employee Performance Analysis

7.1 Sales Performance by Employees

Employee Name	Title	Total Customers	Total Invoices	Total Revenue	Avg/Customer
Jane Peacock	Sales Support Agent	21	212	$1,731.51	$82.45
Margaret Park	Sales Support Agent	20	214	$1,584	$79.20
Steve Johnson	Sales Support Agent	18	188	$1,393.92	$77.44

7.2Top Spending Customer Per Employee

Employee Name	Customer Name	Country	Customer Total Spent
Jane Peacock	Hugh O'Reilly	Ireland	$114.84
Margaret Park	Frantisek Wichterlova	Czech Republic	$144.54
Steve Johnson	Helena Holy	Czech Republic	$128.70

7.3 Employee Hierarchy

Levels	Employee Name	Title
L7	**Madan Mohan**	**Senior General Manager**
L6	Andrew Adams	General Manager
L4	Nancy Edwards	Sales Manager
L3	Michael Mitchell	IT Manager
L2	Robert King	IT Staff
L2	Laura Callahan	IT Staff
L1	Jane Peacock	Sales Support Agent
L1	Margaret Park	Sales Support Agent
L1	Steve Johnson	Sales Support Agent

8. Regional Analysis

8.1 Top and Least Performing Regions

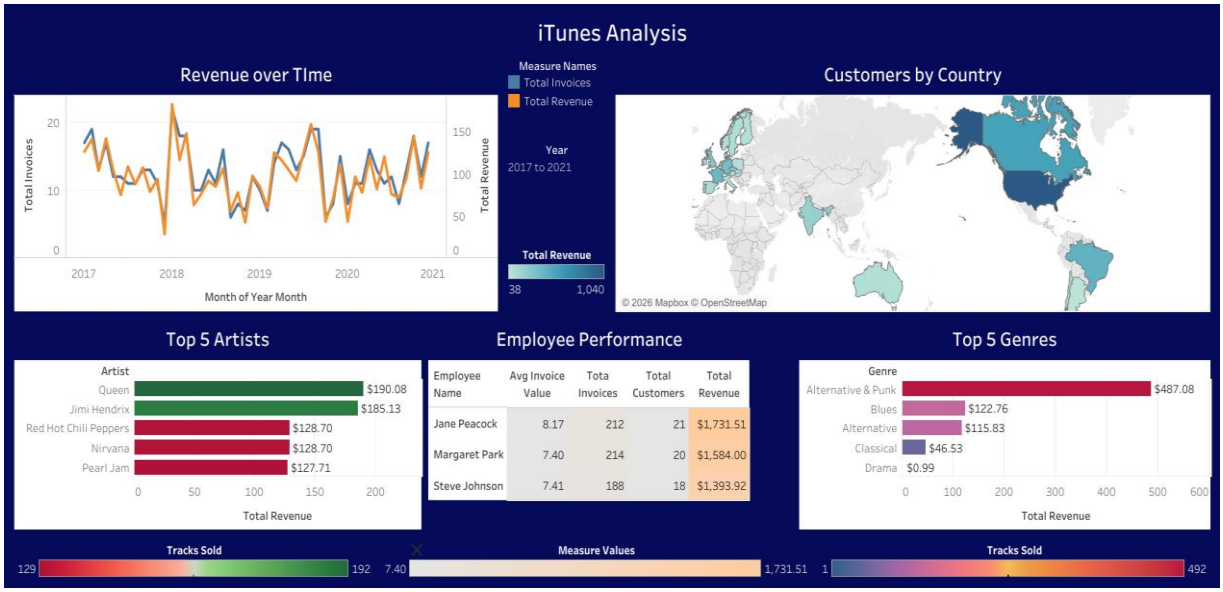
Region	Country	Customers	Revenue	Revenue/Customer
Prague (Best)	**Czech Republic**	**2**	**$273.24**	**$136.62**
Mountain View	USA	2	$169.29	$84.64
London	United Kingdom	2	$166.32	$83.16
Berlin	Germany	2	$158.4	$79.2
Edmonton (Worst)	**Canada**	**1**	**$29.7**	**$29.7**

8.2 Revenue by Country

Country	Total Customers	Total Revenue	Revenue/Customer
USA	13	$1,040.49	$80.04
Canada	8	$535.59	$66.95
Brazil	5	$427.68	$85.54
France	5	$389.07	$77.81
Germany	4	$334.62	$83.66
Czech Republic	2	$273.24	$136.62
United Kingdom	3	$245.52	$81.84
Portugal	2	$185.13	$92.57
India	2	$183.15	$91.57
Ireland	1	$114.84	$114.84

- Denmark was seen to have the lowest total revenue with 1 customer and with a total revenue of $37.62.

9. Dashboard



Dashboard link:

https://public.tableau.com/views/Dashboard_17717710528940/iTunes_Analysis?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

10. Business Recommendations

10.1 Strategy for Genre and Catalog

- Expand the Alternative and Easy listening catalogs as these genres have the highest sales efficiency of 3.44 and 3.08 sales per track.
- Review and run targeted promotions for the zero sales genres or consider removing them from the catalog to reduce cluttering:
    - Bossa Nova
    - Comedy
    - Opera
    - Rock And Rol
    - Sci Fi & Fantasy
    - Science Fiction
    - World
- Reduce investment in the Latin catalog as it has 3.40% of the catalog space but only 3.51% of the total sales ($1.40 sales per track). The Latin genre exhibits the lowest efficiency amongst the active genres.
- Given its global popularity, it's important to look into the Hip Hop/Rap underperformance. It is 0.60% of the catalog share and has a sales per track of $1.57. It shows significant potential for growth.
- Expanding the Argentine catalog with more Alternative & Punk content as it's the only where Rock is not the favourite genre. This shows a unique market opportunity.

10.2 Pricing Strategy

- Consider implementing tiered pricing beyond the $0.99 model.
- Drama and TV Shows use the $1.99 pricing which could be extended to premium content, exclusive releases, or for high demand tracks.
- Since most tracks are priced at $0.99, it is better to focus on strategies to increase purchase volume rather than price increases to grow revenue.
- Bundle pricing can be implemented for frequently purchased track combinations. This can be applied to tracks by Jimi Hendrix as they are commonly bought together. A bundled discount could increase the average invoice value.

10.3 Customer Retention & Reengagement

- Launching a reengagement campaign targeting the 16 inactive customers, prioritizing high value customers like Francois Tremblay, Luis Rojas, and Enrique Munoz.
- Implementing reminder emails every 120-140 days (average purchase gap was 140 days) to encourage customers to visit the store before they become inactive.
- Targeting medium variety customer (5 customers purchasing from 4-6 genres) with genre discovery promotions as higher genre diversity correlates with higher spending.
- Developing loyalty programs for top customers and rewarding frequent buyers as it could increase purchase frequency.

10.4 Sales & Revenue Strategy

- Promotional campaigns in November and December as Q4 performance has an average of only $250.22 when Q1 has an average of $327.44, The holiday season sales promotion campaigns need to be optimized for better performance.
- Maximize marketing for better revenue during peak periods (March and April) with targeted promotions.
- To prevent it from reoccurring, look into the October 2019 revenue crash as there were only 6 invoices with a total revenue of $43.56.
- Monitor the declining average invoice value trend from $7.80 in 2017 to $7.59 in 2020. Even though the difference is small, it reveals that customers are buying fewer tracks per visit over time.

10.5 Regional Expansion

- Focus marketing on Czech Republic customers as they generate the highest revenue per customer of $136.62 and are also the top spending region overall.
- Investigate Denmark's low performance of $37.62 revenue per customer despite 10 invoices.
- Target Ireland, Spain, and Chile for expansion as they have high revenue per customer of $114.84, $98.01, and $97.02 despite having 1 customer each. This shows strong market potential.
- Review the performance of Edmonton, Canada as it's the lowest performing region ($29.70 total revenue with 1 customer). Targeted local promotions may be beneficial.

11. Conclusion

The iTunes store demonstrates a stable and loyal customer base with 100% of the customers being repeat buyers with an average of 10.41 invoices per customer. Rock music was seen to dominate the sales globally, accounting for 55.39% of all purchases. Niche genre opportunities remain largely unexplored.

Revenue has been relatively consistent along the years (2017-2020), ranging between $1.138 and $1.221 per year. Significant month to month volatility and a declining average invoice value indicates areas requiring strategic attention. The business would benefit from targeted reengagement of inactive high value customers, catalog expansion in high efficiency genres, and seasonal promotional campaigns to address weak Q4 performance.

Employee performance is relatively balanced, but workload distribution and incentive structures could be optimised to improve overall performance. Regionally, Czech Republic and Irish customers deliver the highest value per customer and represent models for targeted marketing in underperforming regions.

With strategic implementation of the recommendations outlined in this report, the iTunes music store has significant potential to increase both revenue per customer and overall sales volume across its international customer base.